Noah Mok

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Education

Bachelor of Applied Science, Mechanical Engineering | University of Waterloo | 2026 - 2031

Technical Skills

CAD: **Solidworks**, **Siemens NX**, **AutoCAD**
Coding Languages: **Python** (Sympy, Matplotlib, Numpy, Tkinter), **C++**, HTML, CSS, JavaScript
Technical Knowledge: **Dimensioning and Tolerancing, Solid Modeling, Surface Modeling**, Data Analysis, Statics and Kinematics
Machining and Manufacturing: Drill press, Lathe, Milling machine, Bandsaw, Press Machine
Microsoft 365: Teams, Outlook, Word, Powerpoint, Excel (VBA, Power BI)

Experience

University of Waterloo Automobile Futures | Sep 2026 - active
University of Waterloo

- Lead research on traction capabilities and benchmarking for UWAFT's competition vehicle, the 2026 Jeep Cherokee, and for alternative off-roading vehicle competitors to design traction objectives and optimize vehicle performance.
- Utilized an engineering mindset to design an effective and safe **Rotary Selector Switch Mount** to be integrated into the competition vehicle in 2027.

University of Waterloo Midnight Sun | Sep 2026 - active
University of Waterloo

- Created and manufactured front light mounts using 3D print filament to secure the front lights of the vehicle such that they meet competition constraints and can withstand vibration from the car.
- Designed the **Solar Car Canopy** in **Solidworks** to keep the driver and car components safe, optimize aerodynamic performance, and practice **surface modeling** tools in 3D CAD.

Applied Mathematics Researcher | Jun 2019, Jun 2022
Epsilon Math Camp & MathPath

- Led a team of 16 people in researching isoperimetric sets of English words with Prof. Chip Morgan using **data analysis and Python** to reverse engineer the process behind online tools like auto-correct, published in *Parabola*, Vol. 59, Issue 2.
- Co-developed a solution on Watt's linkage motion with Dr. Zoltan Kovacs, published in the *International Journal of Technology in Mathematics Education*, Vol. 27, No. 4.

Student Page | Sep 2023 – Oct 2025
Edmonton Public Library | Edmonton, AB

- Worked 10-12 hours a week demonstrating a service-oriented mindset, organizing, shelving, and sorting book returns and arrivals and assisting customers in navigating software such as **Word, Powerpoint, and Outlook** .

Projects

Solar Car Canopy | Oct 2026

- Designed a solar car canopy using **Surface Modeling** in **Solidworks** as part of an onboarding challenge for the University of Waterloo Midnight Sun team.
- Crafted an aerodynamic canopy to meet dimensioning constraints, and researched and performed **data analysis** on previous aerobody designs to optimize for canopy drag area to reduce drag by more than **50 percent**.

Rotary Selector Switch and Mount | Sep 2026

- Designed a rotary selector switch and its mount using **Siemens NX** as part of an onboarding challenge for the University of Waterloo Automobile Futures Team.
- Created orthographic views in **AutoCAD** and a design document detailing my solutions to the **dimensioning, safety, and manufacturing** constraints for integration into a production vehicle.
- Improved safety, accessibility, and lifetime of the switch while reducing costs by **20 percent**.

Kinematics Calculator | Jul 2026

- Developed a desktop GUI application through **Python** to solve 1D and 2D kinematic equations to help students with complex physics calculations.
- Utilized **SymPy** to dynamically substitute known values and solve reduced equation systems, saving upwards of **15 minutes per calculation**.

Pendulum Simulation | Jul 2026

- Created a real-time animation of pendulum motion using **Matplotlib** to help students properly visualize simple harmonic motion.
- Embedded pendulum motion equations and learned how different parameters can affect motion through **Python**.
- Implemented an efficient solution in less than 50 lines of code in **O(n) time and O(1) memory**.